

Ahmed Hassan Eldemery

AI Engineer — Machine Learning Engineer

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Professional Summary

AI Engineer and Machine Learning professional specializing in end-to-end ML pipelines, LLM-powered applications, and intelligent automation systems. Skilled in backend API development, computer vision, and prompt engineering using Python, FastAPI, Scikit-learn, OpenCV, and MongoDB. Currently building production-grade systems for multimodal AI classification and business automation at Try GC, delivering measurable impact on real-world operational challenges.

Education

Bachelor of Computer Science (Minor in Artificial Intelligence) Oct 2023 – July 2027 (Expected)

Mansoura University

Mansoura, Egypt

- **Relevant Coursework:** Linear Algebra, Probability & Statistics, Data Structures, Machine Learning, Computer Vision

Technical Skills

- **AI & LLMs:** LLM Integration, Prompt Engineering, LangChain, RAG, Generative AI, Groq API, Llama 3, OpenAI API
- **Machine Learning:** Scikit-learn, XGBoost, Regression, Classification, Clustering, Feature Engineering
- **Computer Vision & Speech:** OpenCV, YOLOv8, PaddleOCR, MediaPipe, TalkNet, Whisper, CLIP
- **Data & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Exploratory Data Analysis (EDA)
- **Backend & Frontend:** FastAPI, Flask, React, REST APIs, WebSockets, JWT Authentication, PGVector, Qdrant
- **Tools & Architecture:** Python, SQL, Git, Docker, Docker Compose, Microservices, Chrome Extensions (MV3)

Professional Experience

AI Specialist

Apr 2026 – Present

Try GC (Formerly Grand Community)

On Site

- Built a multimodal influencer classification system processing **60,000+ profiles**, combining computer vision (face/video analysis) and speech models (TalkNet, Whisper) to auto-classify creators.
- Developed a Manifest V3 Chrome Extension and FastAPI backend to record meetings, transcribe Arabic conversations via Whisper, and extract structured marketing insights using Llama-3.3 via Groq.
- Designed LLM integration pipelines and prompt workflows, reducing manual operational efforts across the company's core database.

Machine Learning Engineer Trainee

June 2025 – Dec 2025

Huawei & Mansoura University Collaboration

Mansoura, Egypt

- Engineered end-to-end ML pipelines (Python, Scikit-learn), improving model accuracy by ~12% via feature engineering.
- Trained and evaluated supervised and unsupervised models on real-world datasets.

Machine Learning Engineer Intern

July 2025 – Aug 2025

National Telecommunication Institute (NTI)

Mansoura, Egypt

- Developed a predictive analytics model using Scikit-learn to forecast e-commerce customer spending behavior.
- Executed end-to-end ML data pipelines, enabling data-driven insights from historical sales data.

Key Projects

Influencer Video Classifier (Try GC)

Python, OpenCV, TalkNet, Whisper

- Engineered a scalable production pipeline combining computer vision with speech models to classify creators based on content type and speaking style.
- Automated marketing operations by fully replacing manual classification workflows across a massive database.

AI Meeting Assistant (Try GC)

FastAPI, Chrome Extension (MV3), Groq API, Whisper

- Engineered a Manifest V3 Chrome Extension utilizing offscreen documents to bypass service worker constraints for real-time audio mixing and secure microphone capture.
- Built a stateless FastAPI backend with strict Pydantic validation to process audio chunks via Groq's Whisper-large-v3 and extract structured Arabic marketing insights using Llama-3.3 in JSON mode.

VoltIQ — Smart Electricity Platform

FastAPI, Flask, YOLOv8, PaddleOCR, XGBoost, PGVector, Docker

- Architected a microservices AI platform combining computer vision OCR, predictive ML, and a RAG document assistant.
- Built an OpenCV/PaddleOCR pipeline to read LCD 7-segment displays and trained XGBoost models for anomaly detection and balance depletion forecasting.

- Deployed a FastAPI RAG microservice utilizing PGVector/Qdrant and LLMs (Gemini/OpenAI) for context-grounded Q&A.

YAQIZ — AI Workplace Safety & Fatigue Monitoring

FastAPI, React, YOLOv8, MediaPipe, Docker

- Built a full-stack platform combining YOLOv8 PPE detection with ByteTrack worker tracking and MediaPipe fatigue analysis.
- Engineered WebSocket pipelines for live video streaming and containerized the full stack using Docker for scalable deployment.

Digital Employee – AI MSME Sales Assistant

Python, Flask, OpenAI API, MongoDB, JWT

- Built a bilingual LLM chat assistant converting conversations into structured orders with auto-generated IDs.
- Implemented role-based access, low-stock alerts, and config-driven workflows for 10+ business sectors.

ErgoAI – Intelligent Health & Productivity Assistant

Python, OpenCV, Computer Vision

- Developed an AI-powered desktop app utilizing OpenCV to detect blink rate, yawning, and posture at ~20 FPS.
- Achieved **3rd Place** in the Computer Vision Course (Mansoura University) and **3rd Place** at the Ibtikar 8 Innovation Competition.

Data & Academic Projects

Scikit-learn, GMM, Pandas

- Developed predictive E-commerce spending models, USA Adult Income EDA, and unsupervised GMM Image Segmentation pipelines.

Hackathons & Competitions

- **Huawei Developer Competition (Northern Africa):** Advanced to Top 50 teams (Team Sanad) presenting an AI-driven solution in a regional competitive environment (2025).
- **First AI Summit Week Hackathon (Mansoura Univ):** Developed and exhibited the initial prototype of the Digital Employee Assistant alongside teams from 16 universities (2026).
- **Egypt Hackathon 4.0 (GDG Delta):** Built the initial prototype of YAQIZ in a highly competitive hackathon among 120+ teams and 600+ developers (2025).

Honors & Certificates

- **Awards:** 3rd Place – Ibtikar 8 Innovation Comp (2025) — 2nd Place – AI Course Comp (Apr 2025) — Digitopia Semi-Finalist (June 2025).
- **ALX Africa Data Science Program:** 15-month intensive track in Data Analytics and Machine Learning (June 2026).
- **Certifications:** NVIDIA (Building LLM Applications, AI for All) — Intro to Machine Learning (Grade: Excellent) — Applied Data Science & Python.